Amirpasha Hedayat

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Research Interests

My research focuses on scientific machine learning and AI for science, with an emphasis on reduced-order models (model reduction), operator learning, and neural approaches for modeling complex dynamical systems. Beyond specific applications, I'm interested in how AI can serve as a general-purpose modeling tool, capable of adapting across scientific domains and levels of abstraction.

Education

PhD **University of Michigan**, Aerospace Engineering & Scientific Computing — Aug 2023 – Present (expected 2028), Ann Arbor, MI, USA

- Also affiliated with Michigan Institute for Computational Discovery and Engineering (MICDE)
- Advisor: Prof. Karthik Duraisamy

MSc **University of British Columbia**, Mechanical Engineering — Aug 2020 – Nov 2022, Vancouver, BC, Canada

- GPA: 4.0/4.0 (92% local; 1st in class)
- Advisor: Prof. Carl Ollivier-Gooch
- Thesis: "Detecting Missing Flow Separation and Predicting Error in Drag using Supervised Machine Learning" [Link]

BS **Amirkabir University of Technology**, Aerospace Engineering — Sept 2015 – Oct 2019, Tehran, Iran

- GPA: 4.0/4.0 (19.0/20.0 local; 1st in class)
- Advisor: Prof. Sahar Noori
- Thesis: "Estimation of Unknown Heat Flux in a 2-D Heat-Sink using Levenberg-Marquardt Inverse Algorithm"

Experience

University of Michigan, Graduate Research Assistant — Aug 2023 – Present, Ann Arbor, MI, USA

- Developing adaptive reduced-order models with continual learning for real-time model updates.
- Building fast ML/AI surrogate models for complex dynamical systems (e.g., weather, plasma).
- Exploring online adaptation of neural operators for improved generalization.

University of British Columbia, Graduate Research Assistant — Aug 2020 – Nov 2022, Vancouver, BC, Canada

- Built modal decomposition workflow to detect missing flow patterns.
- Trained ML classifiers to automatically flag under-resolved CFD simulations.
- Trained ML regressors to predict drag force from reduced flow features.

Amirkabir University of Technology, Undergraduate Research Assistant — Sep 2017 – Oct 2019, Tehran, Iran

- Implemented inverse heat-transfer algorithm to estimate unknown heat flux in a 2D heat sink.
- Performed CFD simulations of flapping airfoils to study unsteady aerodynamic behavior.

Publications

- **Hedayat, A.**, Padovan, A., and Duraisamy, K. "Adaptive Non-Intrusive Reduced-Order Models: Foundations, Design, and Challenges," submitted to *SMO Special Issue on Reduced Order Modeling, Generative AI, and SciML in Digital Twins*, 2026.
- **Hedayat, A.**, and Duraisamy, K., "Attention-Enhanced Convolutional Autoencoder and Structured Delay Embeddings for Weather Prediction," *Preprint*, January 2025. doi: https://doi.org/10.48550/arXiv.2511.12682
- **Hedayat, A.**, and Ollivier-Gooch, C., "Detecting Under-Resolved Flow Physics Using Supervised Machine Learning," *AIAA Journal*, June 2023. doi: https://doi.org/10.2514/1.J062884
- **Hedayat, A.**, and Ollivier-Gooch, C., "Detecting Missing Flow Separation using Supervised Machine Learning," *AIAA SCITECH 2023 Forum*, January 2023. doi: https://doi.org/10.2514/6.2023-1201
- **Hedayat, A.**, "Detecting Missing Flow Separation and Predicting Error in Drag Using Supervised Machine Learning," *University of British Columbia*, 2022. doi: https://doi.org/10.14288/1.0421334

Talks & Posters

- "Non-intrusive Adaptive Reduced-Order Models for Predictive Modeling of Complex Systems," *18th U.S. National Congress on Computational Mechanics (USNCCM18)*, Chicago, IL, July 2025.
- "Linear and Non-linear Reduced Dimensional Manifolds for Global Weather Predictions," *Model Reduction and Surrogate Modeling (MORe2024)*, San Diego, CA, September 2024.
- "Detecting Missing Flow Separation using Supervised Machine Learning," *AIAA SCITECH 2023 Forum*, National Harbor, MD, January 2023.

Skills

Programming Languages: Python, C/C++, Matlab/Octave, Julia
AI/ML Frameworks: Pytorch, JAX, Tensorflow/Keras, Scikit-learn
Development Tools: Git/GitHub, Autotools/Make, Shell Scripting
Operating Systems: Extensive experience with MacOS, Linux (Ubuntu), and Microsoft Windows
Spoken Languages: English (proficient), Persian (native)

Awards & Honors

- Graduate Student Academic Achievement Award - UBC — 2022
- Natural Sciences and Engineering Research Council of Canada (NSERC) Graduate Fellowship — 2020 - 2022
- International Tuition Award - UBC — 2020 - 2022
- Faculty of Applied Science Graduate Award - UBC — 2020 - 2022
- The Distinguished Aerospace Engineering BSc Graduate - AUT — 2019
- Graduated as the Top Student in Undergraduate Studies (out of ∼150) - AUT — 2019
- Iran's National Elites Foundation (INEF) Scholarship — 2018 - 2019
- Department of Aerospace Engineering Academic Excellence Award - AUT — 2016 - 2019